Mary Orrand
ESS 469 Winter 2026
Final Project Panel Image

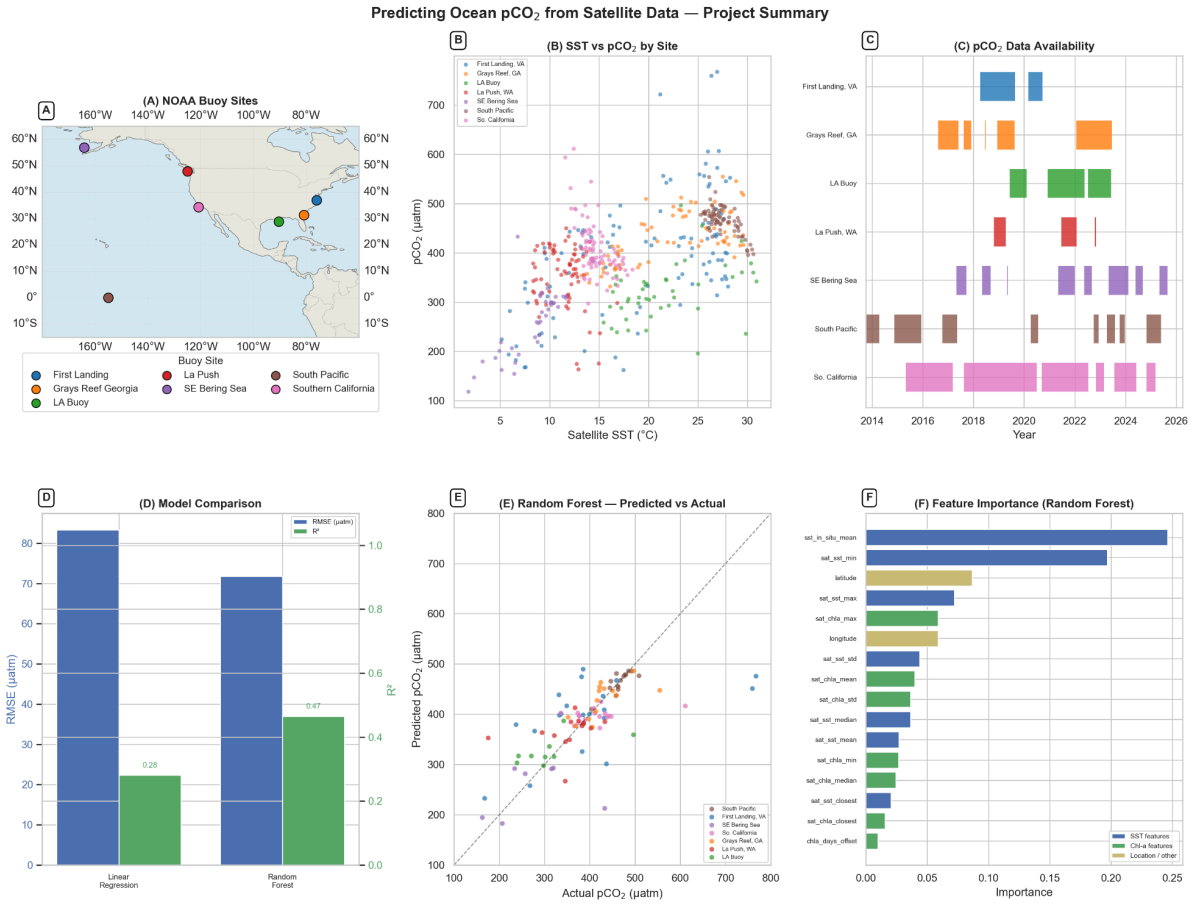

Figure 1. (A) Geographic locations of the 7 NOAA OCADS moored buoy stations used for in-situ pCO□ measurements. (B) Relationship between satellite sea surface temperature and buoy-measured pCO□; non-linear patterns across sites motivate the use of tree-based models. (C) Continuous data availability segments for each buoy site; gaps reflect instrument downtime or data quality exclusions. (D) Model comparison on the held-out test set: RMSE (lower is better) and R² (higher is better); Random Forest significantly outperforms the linear baseline. (E) Test-set predictions vs. actual pCO□ for the best model (Random Forest); points near the 1:1 dashed line indicate accurate predictions. (F) Random Forest feature importance ranked by Gini impurity reduction; SST features dominate, consistent with known CO□ solubility physics.

Figure 1.
(A) Geographic locations of the 7 NOAA OCADS moored buoy stations used for in-situ pCO_2 measurements.
(B) Relationship between satellite sea surface temperature and buoy-measured pCO_2; non-linear patterns across sites motivate the use of tree-based models.
(C) Continuous data availability segments for each buoy site; gaps reflect instrument downtime or data quality exclusions.
(D) Model comparison on the held-out test set: RMSE (lower is better) and R^2 (higher is better); Random Forest significantly outperforms the linear baseline.
(E) Test-set predictions vs. actual pCO_2 for the best model (Random Forest); points near the 1:1 dashed line indicate accurate predictions.
(F) Random Forest feature importance ranked by Gini impurity reduction; SST features dominate, consistent with known CO_2 solubility physics.